SIM1003 Calculus II
Tutorial 8

All even-numbered questions in the following exercises

Exercise 11.1:

Distance

In Exercises 25–30, find the distance between points P_1 and P_2.

25. $P_1(1, 1, 1)$, $P_2(3, 3, 0)$
26. $P_1(-1, 1, 5)$, $P_2(2, 5, 0)$
27. $P_1(1, 4, 5)$, $P_2(4, -2, 7)$
28. $P_1(3, 4, 5)$, $P_2(2, 3, 4)$
29. $P_1(0, 0, 0)$, $P_2(2, -2, -2)$
30. $P_1(5, 3, -2)$, $P_2(0, 0, 0)$

31. Find the distance from the point $(3, -4, 2)$ to the
    (a) xy-plane (b) yz-plane (c) xz-plane
32. Find the distance from the point $(-2, 1, 4)$ to the
    (a) plane $x = 3$ (b) plane $y = -5$ (c) plane $z = -1$
33. Find the distance from the point $(4, 3, 0)$ to the
    (a) x-axis (b) y-axis (c) z-axis
34. Find the distance from
    (a) x-axis to the plane $z = 3$ (b) origin plane $z - x = 2$
    (c) point $(0, 4, 0)$ to the plane $y = x$

Spheres

Find the center C and the radius a for the spheres in Exercises 51–60.

51. $(x + 2)^2 + y^2 + (z - 2)^2 = 8$

52. $(x - 1)^2 + \left(y + \dfrac{1}{2}\right)^2 + (z + 3)^2 = 25$

53. $(x - \sqrt{2})^2 + (y - \sqrt{2})^2 + (z + \sqrt{2})^2 = 2$

54. $x^2 + \left(y + \dfrac{1}{3}\right)^2 + \left(z - \dfrac{1}{3}\right)^2 = \dfrac{16}{9}$

55. $x^2 + y^2 + z^2 + 4x - 4z = 0$

56. $x^2 + y^2 + z^2 - 6y + 8z = 0$

57. $2x^2 + 2y^2 + 2z^2 + x + y + z = 9$

58. $3x^2 + 3y^2 + 3z^2 + 2y - 2z = 9$

59. $x^2 + y^2 + z^2 - 4x + 6y - 10z = 11$

60. $(x - 1)^2 + (y - 2)^2 + (z + 1)^2 = 103 + 2x + 4y - 2z$

Exercise 11.2

Lengths and Direction

In Exercises 25–30, express each vector as a product of its length and direction.

25. $2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$

26. $9\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$

27. $5\mathbf{k}$

28. $\frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{k}$

29. $\frac{1}{\sqrt{6}}\mathbf{i} - \frac{1}{\sqrt{6}}\mathbf{j} - \frac{1}{\sqrt{6}}\mathbf{k}$

30. $\frac{1}{\sqrt{3}}\mathbf{i} + \frac{1}{\sqrt{3}}\mathbf{j} + \frac{1}{\sqrt{3}}\mathbf{k}$

34. Find a vector of magnitude 3 in the direction opposite to the direction of $\frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j} - \frac{1}{2}\mathbf{k}$.

Direction and Midpoints

In Exercises 35–38, find (a) the direction of $\overrightarrow{P_1P_2}$ and (b) the midpoint of line segment P_1P_2.

35. $P_1(-1, 1, 5)$, $P_2(2, 5, 0)$

36. $P_1(1, 4, 5)$, $P_2(4, -2, 7)$

37. $P_1(3, 4, 5)$, $P_2(2, 3, 4)$

38. $P_1(0, 0, 0)$, $P_2(2, -2, -2)$

Exercise 11.6

Matching Equations with Surfaces

In Exercises 1–12, match the equation with the surface it defines. Also identify each surface by type (paraboloid, ellipsoid etc). The surfaces are labelled (a) – (l).

1. $x^2 + y^2 + 4z^2 = 10$

2. $z^2 + 4y^2 - 4x^2 = 4$

3. $9y^2 + z^2 = 16$

4. $y^2 + z^2 = x^2$

5. $x = y^2 - z^2$

6. $x = -y^2 - z^2$

7. $x^2 + 2z^2 = 8$

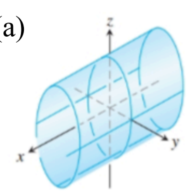
8. $z^2 + x^2 - y^2 = 1$

9. $x = z^2 - y^2$

10. $z = -4x^2 - y^2$

11. $x^2 + 4z^2 = y^2$

12. $9x^2 + 4y^2 + 2z^2 = 36$

(a)


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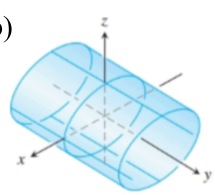

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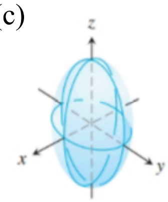

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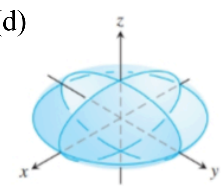

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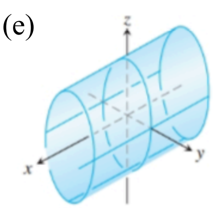

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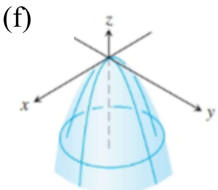

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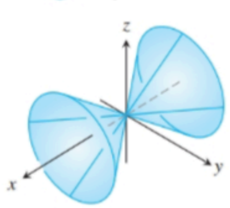

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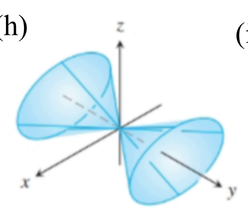

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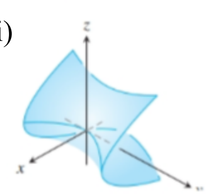

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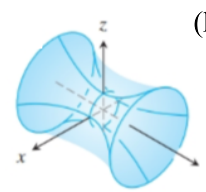

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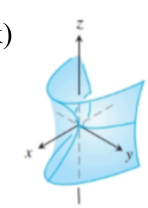

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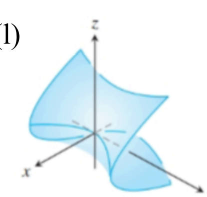

Drawing

Sketch the surfaces in Exercises 13–44.

CYLINDERS

13. $x^2 + y^2 = 4$

14. $z = y^2 - 1$

15. $x^2 + 4z^2 = 16$

16. $4x^2 + y^2 = 36$

ELLIPSOIDS

17. $9x^2 + y^2 + z^2 = 9$

18. $4x^2 + 4y^2 + z^2 = 16$

19. $4x^2 + 9y^2 + 4z^2 = 36$

20. $9x^2 + 4y^2 + 36z^2 = 36$

PARABOLOIDS AND CONES

21. $z = x^2 + 4y^2$

22. $z = 8 - x^2 - y^2$

23. $x = 4 - 4y^2 - z^2$

24. $y = 1 - x^2 - z^2$

25. $x^2 + y^2 = z^2$

26. $4x^2 + 9z^2 = 9y^2$

HYPERBOLOIDS

27. $x^2 + y^2 - z^2 = 1$

28. $y^2 + z^2 - x^2 = 1$

29. $z^2 - x^2 - y^2 = 1$

30. $(y^2/4) - (x^2/4) - z^2 = 1$

HYPERBOLIC PARABOLOIDS

31. $y^2 - x^2 = z$

32. $x^2 - y^2 = z$

ASSORTED

33. $z = 1 + y^2 - x^2$

34. $4x^2 + 4y^2 = z^2$

35. $y = -(x^2 + z^2)$

36. $16x^2 + 4y^2 = 1$

37. $x^2 + y^2 - z^2 = 4$

38. $x^2 + z^2 = y$

39. $x^2 + z^2 = 1$

40. $16y^2 + 9z^2 = 4x^2$

41. $z = -(x^2 + y^2)$

42. $y^2 - x^2 - z^2 = 1$

43. $4y^2 + z^2 - 4x^2 = 4$

44. $x^2 + y^2 = z$